

===== Linux OS =====

=> Where we will use linux os in real-time

- => Jenkins Server
- => Docker Server
- => K8S cluster
- => SonarQube server
- => Nexus Server
- => ELK stack
- => Grafana & Prometheus servers
- => Ansible server

===== What is OS ? =====

=> It is a software which acts as mediator between user and computer.

=> Users will communicate with computers using OS

=> Without OS we can't use any computer

=> OS provides platform to run our applications in computer.

Ex: notepad, calculator, browser, tomcat....

=> We have several operating systems in market

Ex: Windows, Linux, Mac, Android, IOS etc...

===== Windows OS =====

=> Developed by Microsoft company (Bill Gates)

=> Windows os is licensed (commercial)

=> Windows is single user based os

=> Security features are less in windows os (anti virus s/w required)

=> Windows is GUI based (graphical user interface)

=> Windows is recommended for personal use

Ex: watch movies, play games, online classes....

===== Linux OS =====

=> Linux is community based os (not specific to any company)

=> Linux is free and open source os

=> Linux is Multi User based OS

=> Linux is highly secured (anti virus is not required)

=> Linux is CLI based os (command line interface)

=> Linux is highly recommended for business use (servers management)

Ex: App Servers, DB servers, jenkins, docker, k8s, nexus....

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Linux History

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=> Linux OS developed by "Linus Torvalds"

-> Linus Torvalds identified some challenges/issues in Unix OS

-> Linus Torvalds identified one OS which is matching with his ideas

i.e Minux os

-> Linus Torvalds used Minux OS code and made some changes and released into market as new OS i.e Linux OS.

(Li) nus + Mi (nux) = Linux

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Linux Distributions

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-> Linus Torvalds provided Linux OS source code for free of cost

-> So many companies downloaded Linux OS source code and modified according to their requirement and released into market with their brand names

Those are called as Linux Distributions/ Linux Flavours.

-> We have 200+ Linux Distributions in the market.

Ex: Amazon Linux, Ubuntu, Red HAT, suse, debian, kali, fedora....

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How to setup Linux Machine ?

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Approach-1 : Download and Install Linux OS in our System.

Approach-2 : Use Virtual Box and install Linux os as guest os

Approach-3 : Setup Linux VM in AWS Cloud for free of cost

(1 year & monthly 750 hours you can use)

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ðŸ’% *AWS Account Setup* : https://youtu.be/xi-JDeceLeI?si=4MgBX_H4_NPuzdT8

ðŸ’% *Linux VM Setup in AWS* : <https://www.youtube.com/watch?v=Jm1QaTXvw5o>

ðŸ’% *Connect Linux VM with MobaXterm* : <https://youtu.be/uI2iDk8iTps?si=ZuZs0lQTxoRpbRMk>

ðŸ’% *Connect Linux VM with putty* : https://youtu.be/GXc_bxmP0AA?si=HgSydrP89mPxv23s

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Linux Commands

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whoami : display logged in username

pwd : display present working directory

date : display today's date

cal : display current month calendar

cal 2025 : display 2025 year calendar

clear : clear the terminal content

mkdir : make directory (create folder)

 mkdir aws
 mkdir devops
 mkdir docker

rmdir : remove empty directory (delete)

 rmdir devops

rm -rf <dir> : remove non-empty directory

 rm -rf devops

ls : display present working directory content

ls -l : long list the files in alphabetical order

ls -lr : display files in reverse of alphabetical order

ls -lt : display latest files on top

ls -ltr : display old files on top

ls -la : display hidden files

touch : To create empty files in linux

 touch f1.txt
 touch f2.txt f3.txt

cd : change directory

 cd dir-name

 cd ..

rm : to delete file

 rm f1.txt

 rm -rf <dir-name>

mv : For rename & move

 mv existing-name new-name

 mv presentation-location new-location

cat : create new file with data + append data to file + print file data

 # create new file with data

```
cat > f1.txt

# append data to existing file
cat >> f1.txt

# print file data
cat f1.txt

# print file data along with line numbers
cat -n f1.txt
```

tac : To print file data from bottom to top

```
=====
10-July-2024 (Linux Commands)
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```

cp : copy file data from one file to another file

```
cp f1.txt f2.txt
```

Note: If we want to copy data from multiple files then we should use cat command

```
cat f1.txt f2.txt > f3.txt
```

head : print first 10 lines of the file

```
head f1.txt
```

```
head -n 20 f1.txt
```

```
head -n 50 f1.txt
```

tail : print last 10 lines of the file

```
tail f1.txt
```

```
tail -n 25 f1.txt
```

Note: To get latest log message from log file we will use tail command.

grep : grep stands for global regular expression print

Note: Using grep command we can search for content in the file

```
# print lines which contains teen keyword
grep 'teen' abc.txt
```

```
# ignore case sentiveness
grep -i 'teen' abc.txt
```

```
# print lines contains teen with line number
grep -n 'teen' abc.txt
```

```
# print lines which doesn't contain teen keyword
grep -v 'teen' abc.txt
```

```
# search for teen keyword in all files of pwd
grep 'teen' *
```

```
# tail with grep combination
# in last 10 lines of file search for 'teen' keyword
```

```
tail abc.txt | grep 'teen'
```

wc : word count

```
=====
Text Editors in Linux
=====
```

=> vi (visual editor) it is default editor in linux machines

=> Using 'vi' we can create new files and we can modify existing file data.

=> vi command is having 3 modes

1) command mode (just to open the file)

Ex: vi <filename>

2) insert mode (to edit the file) ---> press 'i' in keyboard

3) esc mode (to comeout from insert mode) --> press 'esc' in keyboard

Save changes & close the file => :wq

Without saving changes close the file => :q!

Note: vi command will open the file if it is available otherwise it will create new file and it will open that file.

```
=====
file creation commands
=====
```

touch : to create empty file

cat : create file with data

cp : copy one file data into another file (cp f1.txt f2.txt)

vi : create and open file for editing (vi f3.txt)

```
=====
reading file data
=====
```

cat : print file data from top to bottom

tac : print file data from bottom to top

head : print first 10 lines of file data

tail : print last 10 lines of file data

vi : open the file

```
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```

```
=====
SED Command
=====
```

=> SED stands for stream editor

=> SED is used to process the data (substitute, delete, print, insert)

=> Using SED command we can perform operations on the file without opening the file.

=> SED is very powerful command in linux

```
# replace first occurrence of linux keyword with unix
sed 's/linux/unix/' data.txt
```

```
# replace second occurrence of linux keyword with unix
sed 's/linux/unix/2' data.txt
```

```
# replace 3rd occurrence of linux keyword with unix
sed 's/linux/unix/3' data.txt
```

```
# replace all occurrences of linux keyword with unix
sed 's/linux/unix/g' data.txt
```

```
# substitute and save changes in original file
sed -i 's/linux/unix/g' data.txt
```

```
# delete first line of the file
sed -i '1d' data.txt
```

```
# delete fourth line of the file
sed -i '4d' data.txt
```

```
#delete last line of the file
sed -i '$d' data.txt
```

```
# delete data from nth line to last line
sed -i 'n,$d' data.txt
```

Note: n is a number

```
# delete data from 5th line to 15th line
sed -i '5, 15d' data.txt
```

```
# print data from line number 10 to 20
sed -n '10,20p' data.txt
```

```
# insert data at 2nd line
sed '2i\i love india' data.txt
```

```
# insert data at last line
sed '$a\i am from ashokit' data.txt
```

```
=====
User Management
=====
```

=> Linux is a multi user based OS

=> Multiple users can access single linux machine and can perform multi tasking at time.

Note: "ec2-user" is a default user in amazon linux vm. ec2-user having sudo privileges.

=> Within one linux machine we can create multiple user accounts

=> when we create user account, for user one home directory will be created.

ec2-user => /home/ec2-user

john => /home/john

smith => /home/smith

```
# create user
sudo useradd <uname>

# set password for user & update pwd for user
sudo passwd <uname>

# display all users created
cat /etc/passwd

# swith user account
su <uname>

# navigate to current user home directory
cd ~

# Delete user
$ sudo userdel <uname>

# Delete user along with user home directory
$ sudo userdel <uname> --remove

# how to change username
$ sudo usermod -l <new-name> <old-name>
```

/etc/passwd: Contains general user information.
/etc/shadow: Contains hashed passwords and other security-related information.

```
=====
Working with user groups in linux
=====
```

=> When we create user in linux, for every user one user group also will be created with the given username.

```
# Display all groups in linux
$ cat /etc/group

# Create group in linux
$ sudo groupadd <group-name>

# Adding user to group
$ sudo usermod -aG <group-name> <username>

# Remove user from the group
$ sudo gpasswd -d <username> <group-name>

# display users belongs to a group
$ sudo lid -g <group-name>

# display user belongs to which groups
$ id <username>

# delete group
$ sudo groupdel <group-name>

# changing group name
$ sudo groupmod -n <new-name> <old-name>
```

```
=====
Assignment : create new user and connect with linux vm using newly created user account
=====
```

```
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12-July-2024  
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```

=> In linux, to enable password based authentication we need to modify below 2 files

- 1) sudoers
- 2) sshd_config

```
=====
```

What is sudoers file in Linux

```
=====
```

=> It is very important configuration file in linux machine.

=> Using this file we can control which user can run command as a superuser.

```
# print sudoersfile content  
$ sudo cat /etc/sudeors
```

Note: We should be very careful while working with sudoers file. If we do any mistakes in sudoers file then system will be crashed.

Giving sudo previliges for user

```
# open sudoers file  
$ sudo visudo
```

```
# Add below line  
<username> ALL=(ALL:ALL) ALL
```

=> After making changes to close sudoers file => (CTRL + X + Y + Enter)

```
=====
```

How to enable password based authentication for users ?

```
=====
```

=> In linux vm, by default passwordauthentication is no

=> If we want to connect with linux vm using username and password then we need to set that value as yes.

=> WE WLL MODIFY THIS IN "sshd_config" file.

```
# Display sshd_configurratation file data  
$ sudo cat /etc/ssh/sshd_config
```

```
# Open file  
$ sudo vi /etc/ssh/sshd_config
```

Note: Go to insert mode and enable pwdbasedauthentication as yes

```
# restart sshd service  
# sudo systemctl restart sshd
```

Note : Now we can connect with linux vm using username and pwd

```
$ ssh username@public-ip
```

```
=====
```

File Permissions in Linux

```
=====
```

=> Using file permissions we can secure our files and we can protect our file data.

=> We have 3 types of permissions in linux

r => read

w => write

x => execute

=> file/directory permissions will be represented like below

rw-rw-rw- f1.txt

=> file permissions contains in 9 characters

first 3 characters => user/owner permissions

middle 3 characters => group permissions

last 3 characters => others permissions

r--r-xr-- f2.txt

user : read
group : read + execute
others : read

r-xrw--w- f3.txt

user: read + execute
group: read + write
others : write

=> To change file/directory permissions we will use 'chmod' command

+ => add

- => remove

Giving execute permission for user
\$ chmod u+x f1.txt

giving write permission for group
\$ chmod g+w f1.txt

Remove execute permission for others
\$ chmod o-x f1.txt

Remove all permissions for others
\$ chmod o-rwx f1.txt

give all permissions for group
\$ chmod g+rwx f1.txt

=====
File Permissions in Numeric Format
=====

0 => no permissions

1 => Execute

2 => Write

3 => (2+1) => Write + Execute

4 => Read

5 => (4 + 1) => Read + Execute

6 => (4+2) => Read + Write

7 => (6+1) => Read + Write + Execute

```
# ugo+x
$ chmod 111 f1.txt
```

```
# ugo+w
$ chmod 222 f1.txt
```

```
# u+rx, g+rw o+rx
$ chmod 765 f1.txt
```

```
# u+r, g+rx, o+rw
$ chmod 456 f1.txt
```

```
# u+rx, g+rx, o+rx
$ chmod 777 f1.txt
```

```
# u-rwx, g-rwx, o+rwx
$ chmod 7 f1.txt
```

```
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15-July-2024
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```

```
=====
chown command
=====
```

=> It is used to change file/directory ownership

```
# change owner
sudo chown new-owner file/directory
```

```
# change owner-group
sudo chown :new-group file/directory
```

```
# change owner & group of file/directory
sudo chown new-owner:new-group file/directory
```

```
=====
Q) What is the diff between chmod & chown ?
=====
```

chmod => To change file/directory permissions

chown => To change owner/group

```
=====
How to find the location of files in linux?
=====
```

=> in linux we can use 'find' command to search file paths.

```
# search for the files which are having name as f1.txt
sudo find /home -name f1.txt
```

```
# search for empty files inside /home
sudo find /home -type f -empty

# search for empty directories inside /home
sudo find /home -type d -empty

# print 30 days old files in linux vm
sudo find /home -mtime 30 -print

# print 1 day old files in linux vm
sudo find /home -mtime 1 -print

# delete 30 days old files in linux vm
sudo find /home -mtime 30 -delete
```

```
=====
Working with Zip files in linux
=====
```

```
=> Zip is used for files archieve (compress)

## syntax to create zip file : $ zip <zip-file-name> <content>

# create some empty files
touch f1.txt f2.txt f3.txt

# create zip file with content
zip ashokit_data *.txt

# unzip the zip file
unzip ashokit_data.zip
```

```
=====
Networking commands
=====
```

```
ping : To check connectivity
        ping 192.168.23.321

wget : It is used to download files from internet
        wget <url>

curl : It is used to send HTTP request to server
        curl https://type.fit/api/quotes

ifconfig : To get ip address of the machine
```

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Date : 16-July-2024
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```

```
=====
Package Managers in Linux
=====
```

```
=> package means a software

        Ex: git, maven, java, python etc...

=> Package Managers are used to install / update / manage software packages in linux machines.
```

=> Package managers are specific to linux distribution.

amazon linux / Red Hat Linux / Cent OS : yum

ubuntu / debian : apt

```
# check git client installed or not
git -version
```

```
# install git client s/w
sudo yum install git
```

```
# check git installation path
whereis git
```

```
# check java version
java -version
```

```
# check java path
whereis java
```

```
# install java
sudo yum install java
```

```
=====
Assignment : Remove git and java from linux machine
=====
```

```
=====
Webserver Setup in Linux VM
=====
```

=> Webserver is a software which is used to run websites.

=> Website means collection of web pages

ex: login page, register page, dashboard page, about us page

=> Websites are divided into 2 types

- 1) Static website
- 2) Dynamic website

=> The website which gives same response for every user is called as static website.

=> The website which gives response based on logged in user is called as dynamic website.

=> To run static websites we can use 'httpd' as webserver.

=> To run dynamic websites we can use 'tomcat, iis' as webserver.

```
# install httpd webserver
sudo yum install httpd -y
```

```
# check webserver status
sudo service httpd status
```

```
# start httpd server
sudo service httpd start
```

Note: httpd webserver runs on http protocol with 80 port number.

Note: To access our webserver we need to enable http protocol in ec2 vm security group

inbound rules.

=> Once http protocol enabled we can access our webserver using ec2-linux-vm public ip

```
# navigate to website content directory
cd /var/www/html
```

```
# create a file with a name : index.html and write the content
sudo vi index.html
```

```
=====
Static website hostig in linux
=====
```

```
# install webserver
$ sudo yum install httpd
```

```
# start webserver
$ sudo service httpd start
```

```
# Navigate to website content directory
$ cd /var/www/html
```

```
# create index.html file with website content
$ sudo vi index.html
```

Note: httpd webserver runs on 80 port number.

=> To access our webserver we need to enable 80 port number in security group inbound rules.

=> We can access our webserver using ec2-vm public ip.

```
=====
What is systemctl in linux ?
=====
```

=> systemctl is used to manage services in linux machines.

=> using systemctl we can perform below operations

- a) start a service
- b) stop a service
- c) restart a service
- d) enable/disable service

```
# stop httpd server
sudo systemctl stop httpd
```

```
# start httpd server
sudo systemctl start httpd
```

```
# reload service
sudo systemctl reload http
```

```
=====
```

uptime : from when our linux vm is running

free : to display memory details

top : display running processes

htop : display running process in table format

Q) How to check linux os version ?

Ans) cat /etc/os-release

Q) How to check linux kernel version ?

Ans) uname -r

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                                Date : 18-July-2024  
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```

```
=====
```

Working with Link Files

```
=====
```

=> In linux we can create link files (similar to shorcut files in windows)

=> We have 2 types of link files in linux

- 1) Hard link
- 2) Soft link (symbolic link)

```
-----
```

Syntax To create Hard Link

```
-----
```

```
$ ln <orginal-file> <link-file>
```

```
$ touch f1.txt
```

```
# create link file  
$ ln f1.txt f100.txt
```

Note: f100.txt is link file for f1.txt

```
# print files along with inode numbers  
# ls -li
```

Note: If we write some data to original file, it is reflecting in link file also.

Note: when we delete original file, link file is not effected (hard link).

```
-----
```

Syntax To create Soft Link

```
-----
```

```
$ ln -s <orginal-file> <soft-link-file>
```

Ex:

```
$ touch s1.txt
```

```
$ ln -s s1.txt s100.txt
```

```
$ ls -li
```

Note: Original file and link file having diff inode numbes

```
$ cat >> s1.txt
```

Note: Original file data reflecting in link file also.

```
$ rm s1.txt
```

Note: When we remove original file then soft link file will become dangling file.
We can't access that file.

```
=====
process management
=====
```

Process management in Linux involves controlling and monitoring the execution of programs (processes) on the system.

```
# display processing running
$ ps aux
```

Note: Every process will process id (PID)

```
$ kill PID
```

```
# terminate process immediatley (forcefully)
$ kill -9 PID
```

```
=====
How to change hostname in vm (temporarily) ?
=====
```

```
# set hostname
$ sudo hostname <new-name>
```

```
# re-start session
$ exit
```

Note: connect back to vm then we can see configured hostname

```
=====
How to set hostname permanentley
=====
```

```
# update hostname in below file
$ sudo vi /etc/hostname
```

```
# restart the vm
```

Note: After restart hostname configured in file will be reflected in terminal.

```
=====
Summary
=====
```

- 1) What is Linux OS
- 2) Windows Vs Linux
- 3) Linux VM Setup in AWS cloud
- 4) Connecting with Linux VM using SSH client
- 5) Directories & Files Based Operations
- 6) Text Editors
- 7) Text filters
- 8) User & Group Management
- 9) File Permissions
- 10) File Ownership
- 11) Archieves
- 12) Networking
- 13) suderos file
- 14) sshd_config

- 15) package managers
- 16) static website hosting (httpd)
- 17) Service Management (systemctl)
- 18) Process management
- 19) Working with link files
- 20) Changing Hostname